Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Scam Shield

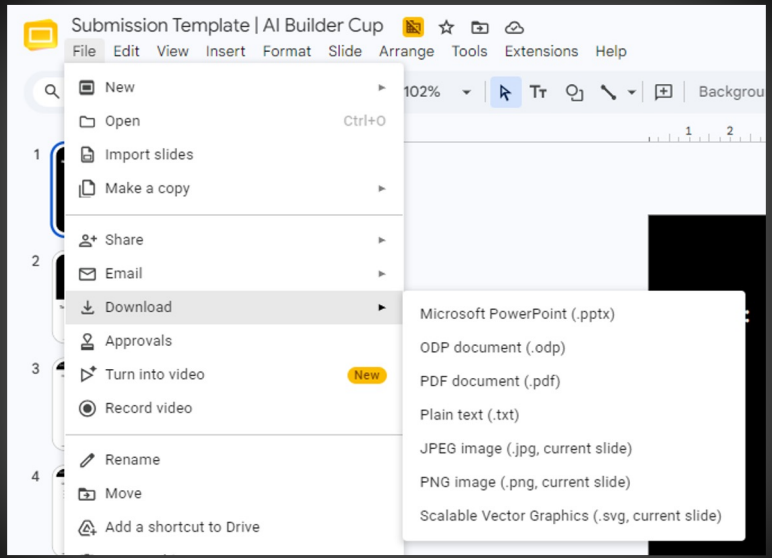
a. Team name: **Scam Shield**

b. Team leader name: **Naman Shrimal**

c. Problem Statement:

BFSI: Intelligent Risk, Fraud & Financial Experiences

People in India lose crores to scam messages, QR codes and calls that look real. Scam Shield lets anyone check what they were sent in seconds, then act: a complaint draft, a shared scam list that warns the next person, and a WhatsApp card for the family. Community benefit: fewer people lose their savings, and every report protects the next one.



Note: Please download this template to make your edits and submit the final version.

To download, go to File > Download > Microsoft PowerPoint (.PPTX). Kindly refrain from sending edit requests.

Brief about the idea

Check what you were just sent before you act on it.

Free for anyone in India; every report warns the next person.

One multimodal Gemini call

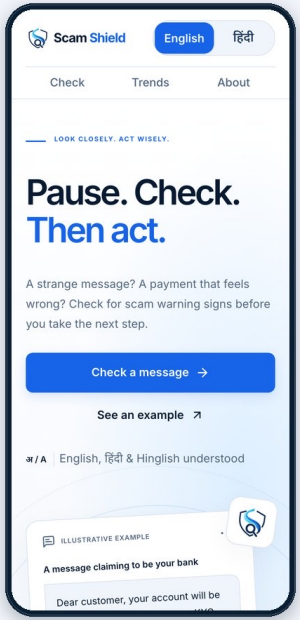
Text, screenshots, QR codes, links and Hinglish voice notes are read together by Gemini on Vertex AI.

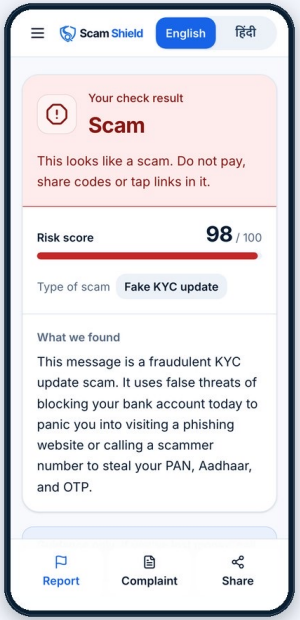
A verdict you can check

Scam, Suspicious or No scam signs found. Every red flag quotes the exact words. English or Hindi.

From verdict to action

A complaint draft for 1930 and cybercrime.gov.in, a shared scam list and a WhatsApp warning card.





The live app. Links are opened in an isolated browser, never on your phone.

Opportunities

Rs 19,813 crore reported lost to cyber fraud in 2025, across about 21.78 lakh complaints (I4C data, National Cyber Crime Reporting Portal)

How different is it from existing ideas?

Truecaller Scam Checker (Sep 2026): checks links, numbers and messages against community reports; screenshot upload is only planned.

Google scam detection: Messages on Pixel 6 and newer, and calls on Pixel 9 and newer, in India.

Chakshu (Sanchar Saathi): a reporting facility for suspicious calls, SMS and WhatsApp messages.

Scam Shield: reads the content itself, so it can flag scams nobody has reported yet, from text, screenshots, QR codes, links and voice notes.

How will it solve the problem?

- Checks before you pay, scan or click
- Quotes every red flag from the message
- Turns a scam into action: complaint, shared list, family warning

USP of the proposed solution

- Catches scams nobody has reported yet
- UPI-aware: a QR that asks you to pay while the text says receive
- Understands Hinglish voice notes

List of features offered by the solution

5 input types
Text, screenshot, QR code, link and voice note.

Clear verdict
Scam, Suspicious or No scam signs found, with a risk score.

Evidence quoted
Each red flag points to the exact words in the message.

English + Hindi
Hinglish understood; answers in the language you pick.

Complaint draft
Ready text for 1930 and cybercrime.gov.in.

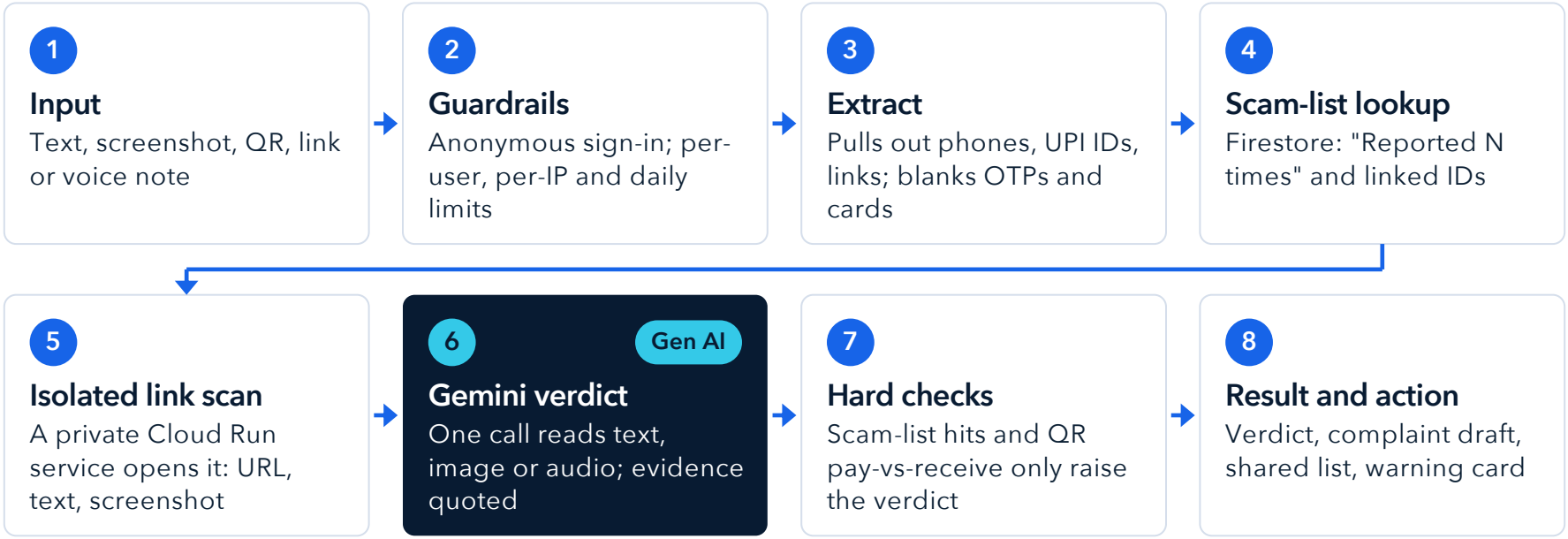
Shared scam list
"Reported N times", with linked numbers, UPI IDs and sites.

Warning card
One tap shares a warning with the family WhatsApp group.

Agent tools
WebMCP tools let AI agents check and report.

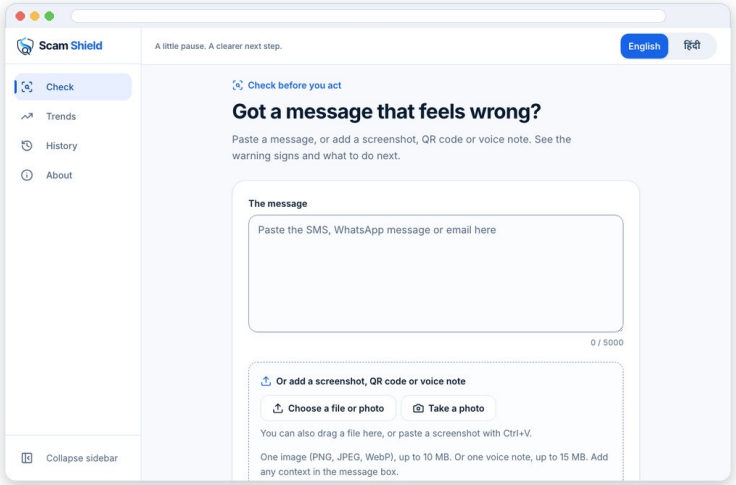
Also: trends, history, one-tap examples, an installable app and the Android share menu.

Process flow diagram or Use-case diagram

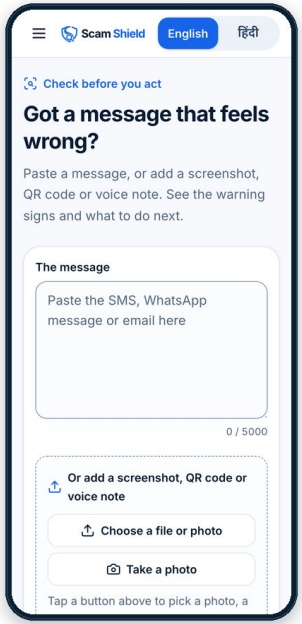


Why Gen AI: rules cannot read a screenshot or a Hinglish voice note, or judge a scam nobody has reported. Gemini does. The hard checks can only raise its verdict, never lower it.

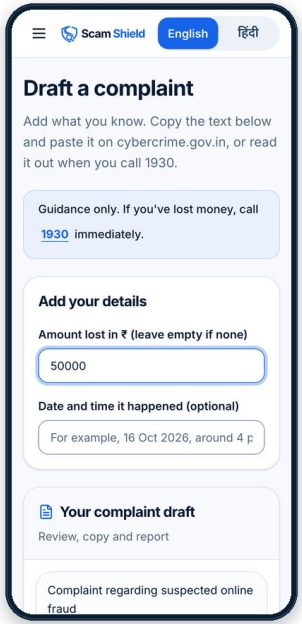
Wireframes/Mock diagrams of the proposed solution (optional)



Web app
The same flow on desktop, with a sidebar.



1 Paste or share
Text, photo, QR or voice note.

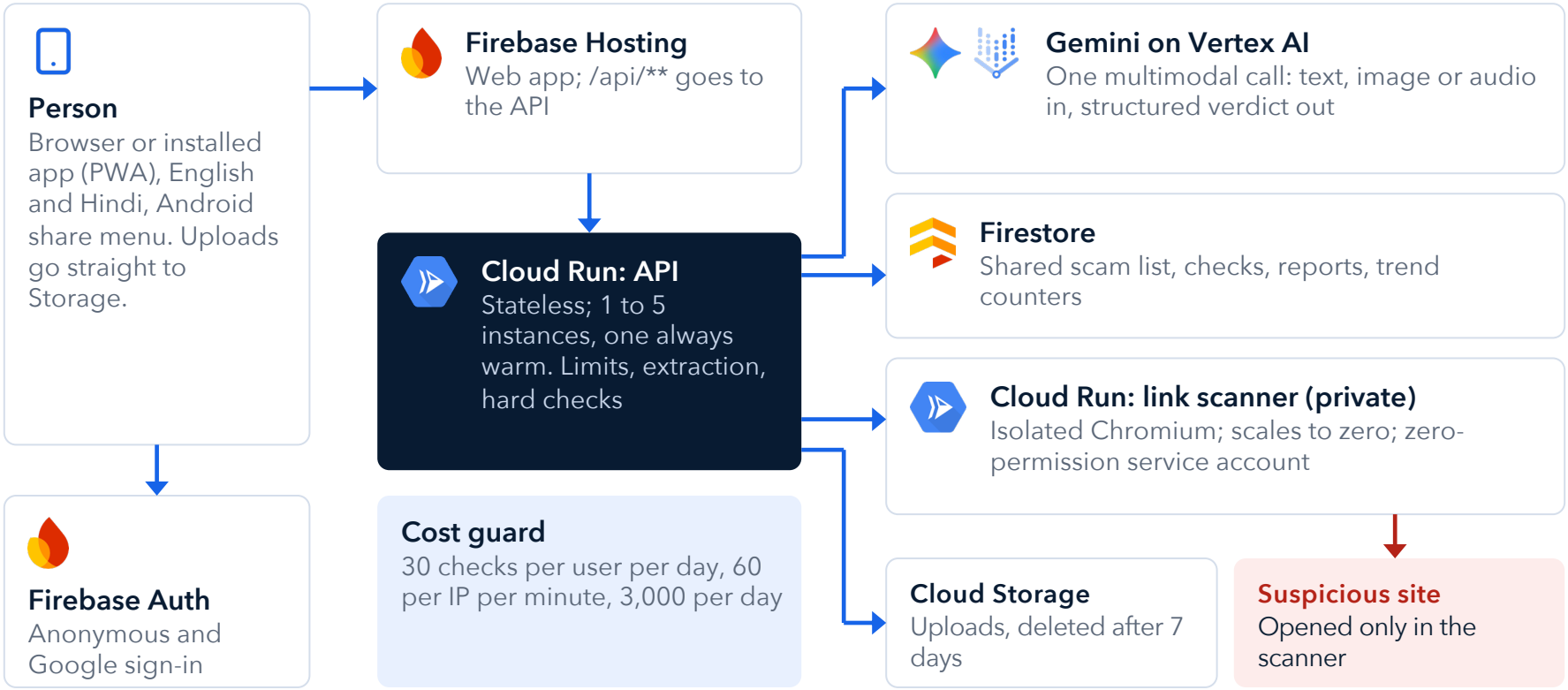


2 Act on a scam
A complaint draft ready to file.



3 Hindi
The whole app switches language.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Gen AI that does real work

- **Gemini 3.7 Flash** on Vertex AI, with Gemini 2.5 Flash as automatic fallback
- **One multimodal call** reads text, screenshots and audio (Hinglish voice)
- **Schema-validated output;** every red flag quotes evidence; English or Hindi
- **Hard checks only raise** the verdict: scam list, QR pay-vs-receive
- **Does what rules cannot:** unreported scams, images, voice
- **WebMCP agent tools:** check_message and report_scam

Google Cloud and Firebase, built to scale

- **Cloud Run API:** stateless, autoscaling, one warm instance
- **Link scanner:** scales to zero, own zero-permission service account
- **Firestore** shared scam list; **Firebase** Hosting, Auth, Storage

Application

React 18 | Vite | Tailwind | Workbox PWA | Node 22 | TypeScript | Hono | zod | Playwright | sharp + jsQR

590+ automated tests, plus a 50-case live benchmark

Estimated implementation cost (optional)

FIXED

about US$13

per month (estimate)

One Cloud Run API instance kept warm: 1 vCPU, 1 GiB, Mumbai. Idle rate US$0.0000025 per vCPU-second and per GiB-second, before any free tier.

PER CHECK

US$0.004–0.008

per text check (estimate)

Gemini 3.7 Flash, assuming about 2,500 input and 500 to 1,500 output tokens. US$0.75 in, US$3.75 out per 1M tokens until 31 Dec 2026; double from 1 Jan 2027.

FREE TIERS

Free tiers

expected at this volume (estimate)

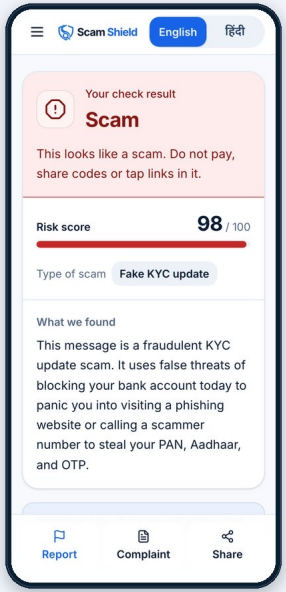
Firebase Hosting, Auth, Firestore and Storage are expected to stay inside free tiers. The link scanner bills only while it scans and scales to zero.

Limits bound the bill
30 checks per user per day, 60 per IP per minute, 3,000 per day overall.

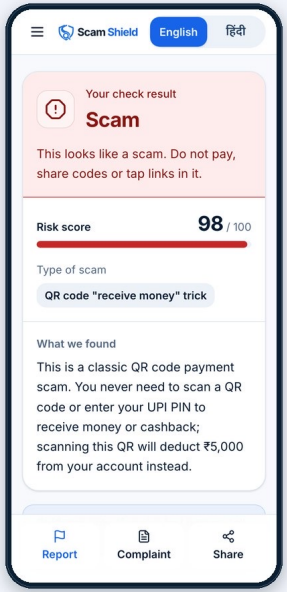
Ceiling at the daily cap
3,000 checks at the per-check estimate is about US$11 to 23 per day for Gemini.

Budget alert
Set: Rs 1,000 per month on this project, alerts at 50%, 90% and 100% of the budget. A hard daily cap also applies.

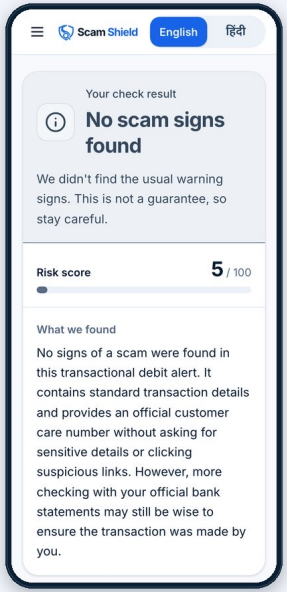
Snapshots of the prototype



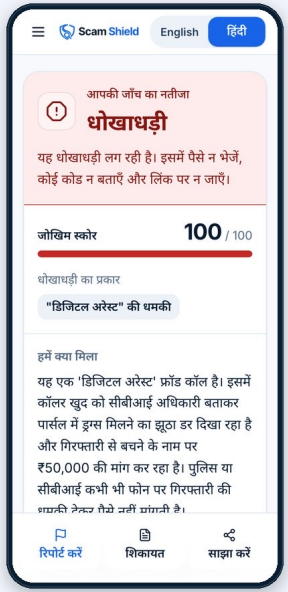
Scam
Fake KYC SMS, evidence quoted



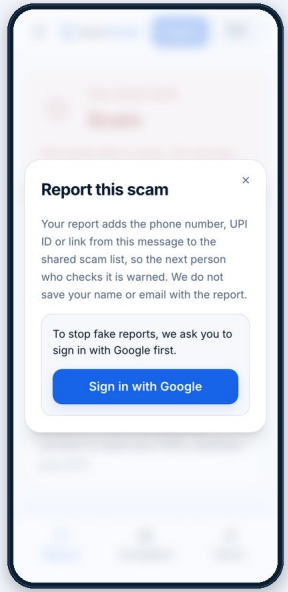
Scam
QR asks you to pay, text says receive



No scam signs found
A genuine bank alert; not a guarantee



Scam, in Hindi
Hinglish voice note, fake police call



Report
Adds it to the shared scam list

Prototype Performance report/Benchmarking

100.0%	100.0%	0.0%	100.0%
Accuracy	Scam recall	False-positive rate	Scam type correct

English run, 2026-10-17, 50 synthetic test cases, live app

Input	Cases	English p50	English p95	Hindi p50	Hindi p95
Text	45	4.5 s	20.8 s	5.0 s	8.6 s
Image	4	7.3 s	8.7 s	8.0 s	9.7 s
Audio	1	10.3 s	10.3 s	11.6 s	11.6 s

Hindi run: accuracy 100.0%, scam recall 100.0%, false positives 0.0%, scam type correct 100.0%; answers returned in Hindi.

p50 is the median time, p95 the slowest 5%, in seconds from request to answer on the live app. The image and voice samples are very small (4 and 1 cases).

Read this honestly

- **50 synthetic test cases, a small test, not a guarantee.** 26 scams across 13 types and 24 genuine messages.
- **Model:** gemini-3.7-flash, with gemini-2.5-flash as fallback.
- Errors: 0. False positives: none.
- **The first run found a real bug** (the model did not know today's date); it was fixed and re-measured.

Additional Details/Future Development (if any)

More Indian languages

Tamil, Telugu, Marathi, Bengali and more.

Bank & telecom partners

Verified reports feed the shared scam list.

URL reputation feeds

Add Google Web Risk next to the isolated scan.

Android app

Share SMS and notifications straight to a check.

Google Search grounding

Switch on for link checks once quota allows.

Call-recording upload

Extend today's voice-note check to recorded calls.

How it scales

Stateless API
Cloud Run adds instances (1 to 5 today) as traffic grows; the daily cap is a setting, not a rebuild.

Managed data
Firestore, Hosting, Auth and Storage need no servers to run; uploads auto-delete after 7 days.

Cost stays bounded
Per-user, per-IP and global limits rise only as budget allows; App Check adds abuse protection next.

Provide links to your:

- **GitHub Public Repository**
  https://github.com/Namans12/scam-shield

- **Demo Video Link (3 Minutes)**
  [add the public video link here] *(owner to fill in)*

- **Final Product Link**
  https://project-91260b43-6705-4176-b71.web.app

Scan to try the live app

Paste a suspicious message and see the verdict.



Scam Shield

Pause. Check. Then act.

Check what you were just sent before you act on it. Free, in English and Hindi.

- Gemini reads any input
- Evidence for every flag
- A scam becomes action

project-91260b43-6705-4176-b71.web.app

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

